ASSIGNMENT 1 REPORT
CO3133 - DEEP LEARNING AND ITS APPLICATIONS

FOUNDATIONS OF DEEP
LEARNING PIPELINES AND ARCHITECTURES: FROM LINEAR MODELS TO
MODERN SEQUENCE MODELS AND TRANSFORMERS

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PROBLEM AND DATA DESCRIPTION

FASHION **IMAGE CLASSIFICATION**: AUTOMATICALLY CATEGORIZE CLOTHING IMAGES FOR RETAIL AND INVENTORY MANAGEMENT.

We conduct all primary experiments on the **Fashion-MNIST** benchmark datase.

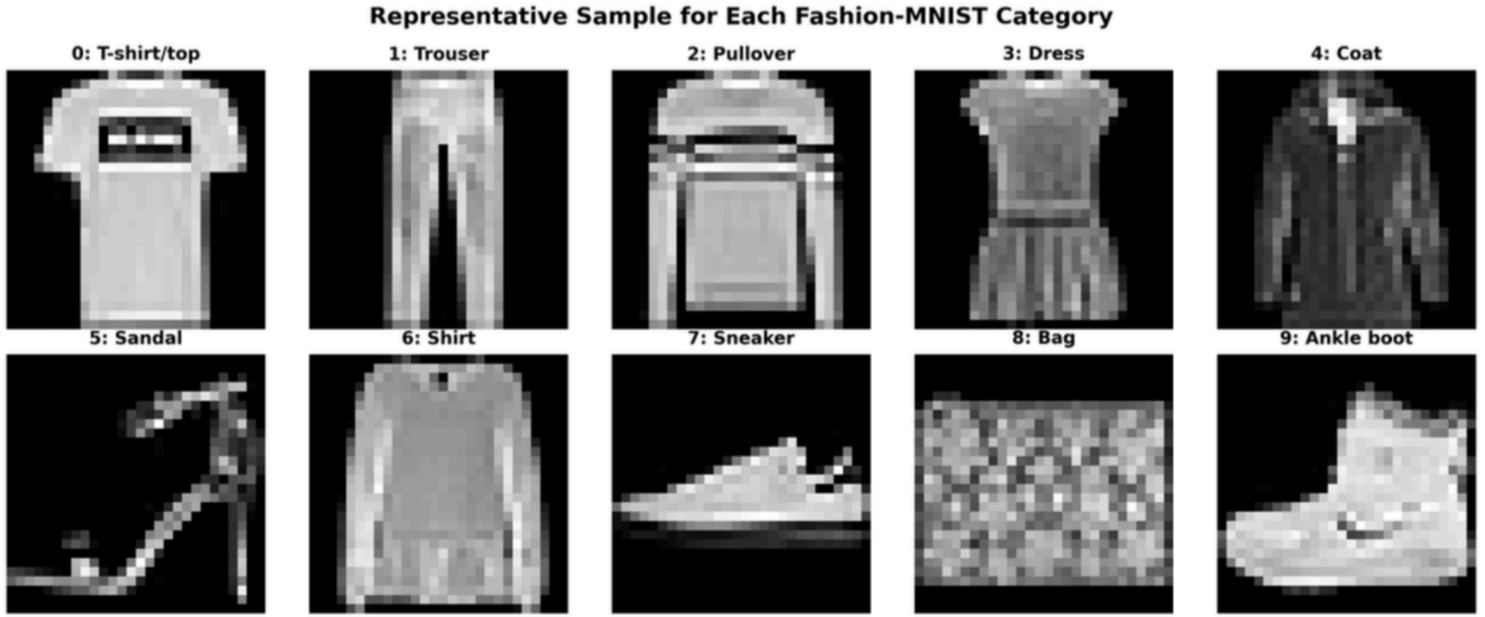


Figure 1.2: Representative samples randomly drawn from each of the ten fashion categories.

- Source: Zalando Research (MIT License)
- Total Samples: **70,000 images**
- Split: **60,000** training / **10,000** testing

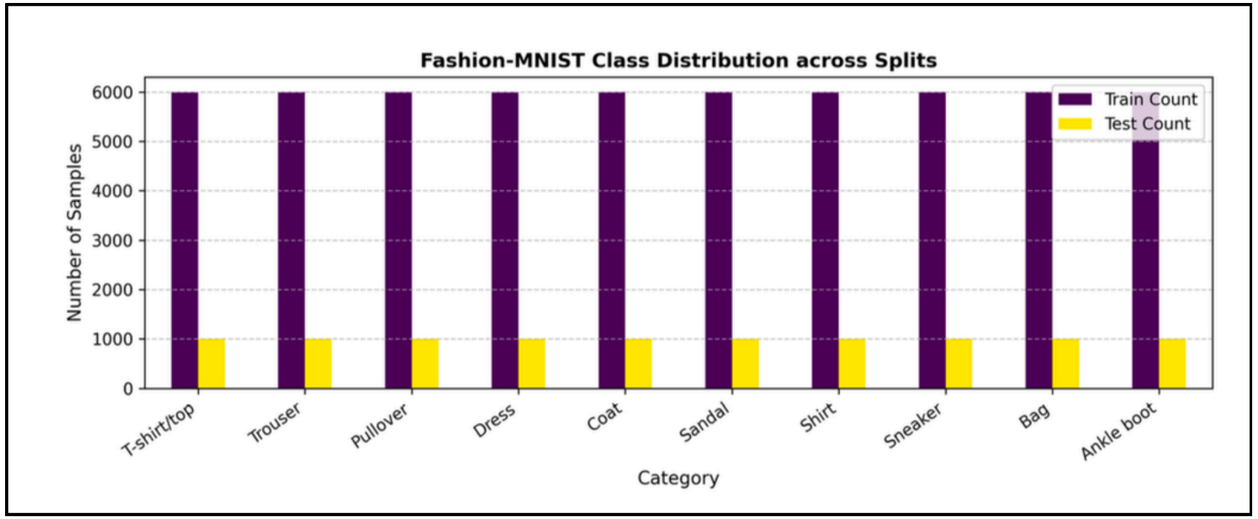
- Image Format: **28 × 28 pixels**, grayscale

- Classes: 10 fashion categories

- Class Distribution: Perfectly balanced (6,000 train + 1,000 test per class)

- Limitations: Low resolution, centered objects, simple backgrounds, limited real-world variability.

EXPLORATORY DATA ANALYSIS (EDA)



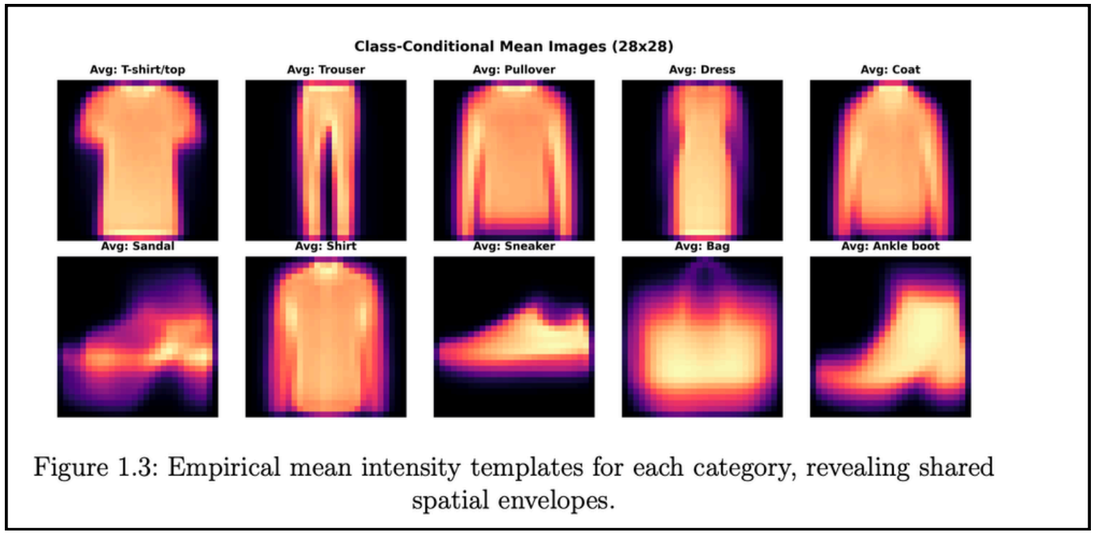


Figure 1.3: Empirical mean intensity templates for each category, revealing shared spatial envelopes.

Data Splitting

1. Training Partition (Dtrain): 54,000 samples (90.0% of the official training set).

2. Validation Partition (Dval): 6,000 samples (10.0% of the official training set).

3. Held-out Test Partition (Dtest): 10,000 samples (100.0% of the official test set).

METHOD & EXPERIMENTAL SETUP

PIPELINE OVERVIEW

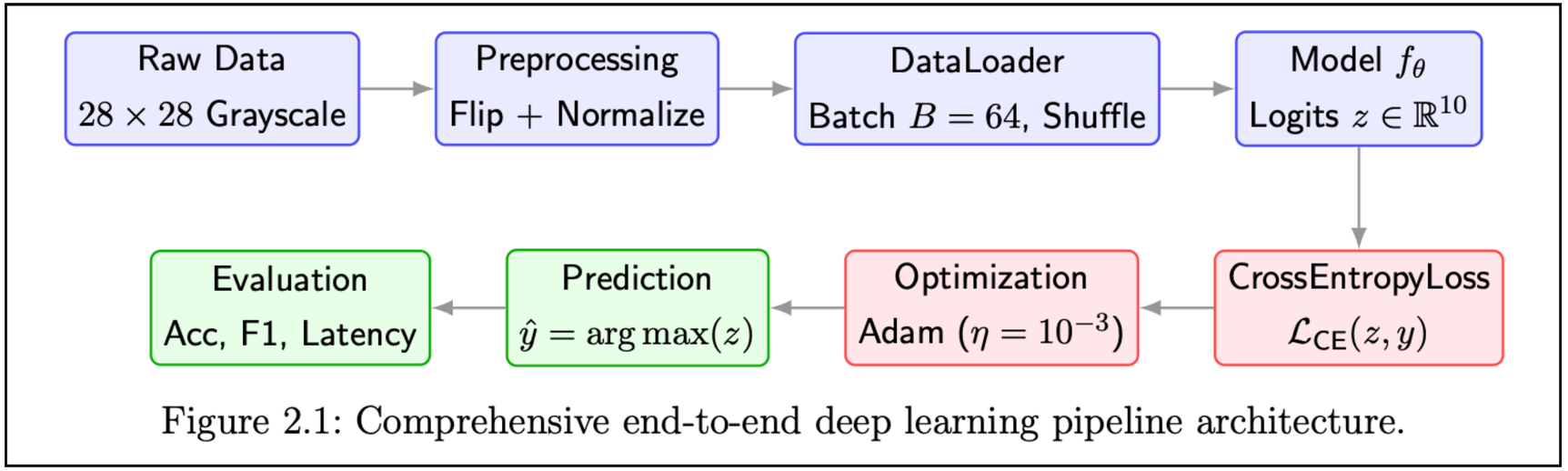


Figure 2.1: Comprehensive end-to-end deep learning pipeline architecture.

T*raining Setup:*

- Optimizer: Adam (LR = 1e-3), Adam (β1 = 0.9,β2 = 0.999,ϵ = 10−8)
- Regularization: Weight Decay = 1e-4, Dropout = 0.1–0.3
- Batch Size: 64
- Epochs: 20
- Checkpoint: Lowest Validation Loss
- Hardware: NVIDIA Tesla T4 (16 GB VRAM) - Google Colab
- Framework: PyTorch (FP32 Precision)
- Fair Comparison: Identical training configurations across all models.

MODEL ARCHITECTURES

01 / Linear Classifier

Input (784) → Linear (10) → Logits

02 / Multilayer Perceptron (MLP)

Input (784) → FC (256) → ReLU → Dropout → FC(128) → ReLU → Dropout → FC(10)

03 / Convolutional Neural Network (CNN)

Input (1×28×28) → [Conv 3×3 → BN → ReLU → MaxPool] × 2 → Flatten → FC (128) → ReLU → Dropout → FC (10)

Conv channels: 1 → 32 → 64

04 / Recurrent Models (LSTM / GRU)

Input (28×28) → 28 Sequential Rows → LSTM / GRU → Final Hidden State → FC (10)

Baseline: H = 128 | Matched Budget: LSTM H = 258, GRU H = 300

05 / Vision Transformer (ViT)

Input (28×28) → Patch Embedding (4×4) → 49 Tokens + [CLS] → 2 Transformer Encoders → FC (10)

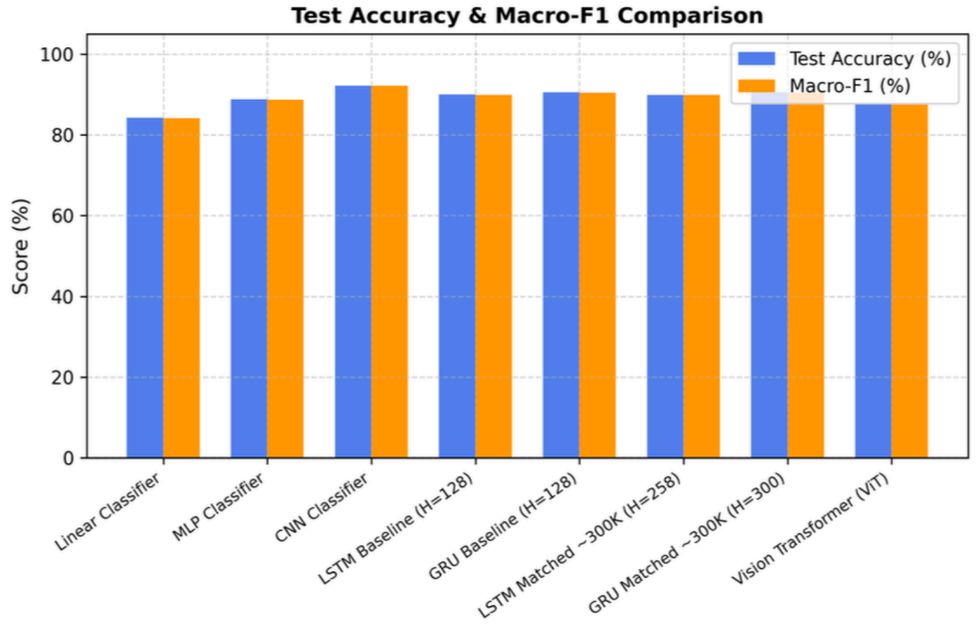
Embedding Dim = 64 | Attention Heads = 4 | Trained from Scratch

MAIN EXPERIMENTAL RESULTS

Table 3.1: Comprehensive Performance and Computational Complexity Comparison on Fashion-MNIST.

Model	Params	Epoch	Val Loss	Val Acc	Test Acc	Macro-F1	Train (s)	Latency
Linear Classifier	7,850	12	0.4352	85.13%	84.21%	84.19%	276.4 s	0.0027 ms
MLP Classifier	235,146	17	0.3026	88.98%	88.77%	88.71%	285.8 s	0.0121 ms
★ CNN Classifier	421,834	9	**0.1994**	**92.63%**	**92.19%**	**92.17%**	306.7 s	0.0205 ms
Recurrent Models — Baseline ($H = 128$):								
LSTM Baseline ($H = 128$)	82,186	17	0.2676	90.33%	90.00%	89.95%	289.1 s	0.0242 ms
GRU Baseline ($H = 128$)	61,962	17	0.2582	90.80%	90.51%	90.46%	292.6 s	0.0202 ms
Recurrent Models — Matched Parameter Budget ($\approx 300K$):								
LSTM Matched ($H = 258$)	299,806	14	0.2591	90.80%	89.94%	89.91%	305.5 s	0.0475 ms
★ GRU Matched ($H = 300$)	300,010	10	0.2532	91.02%	90.60%	90.49%	303.9 s	0.0460 ms
Vision Transformer (ViT)	72,074	20	0.3170	88.07%	87.60%	87.52%	373.5 s	0.0802 ms

MAIN EXPERIMENTAL RESULTS



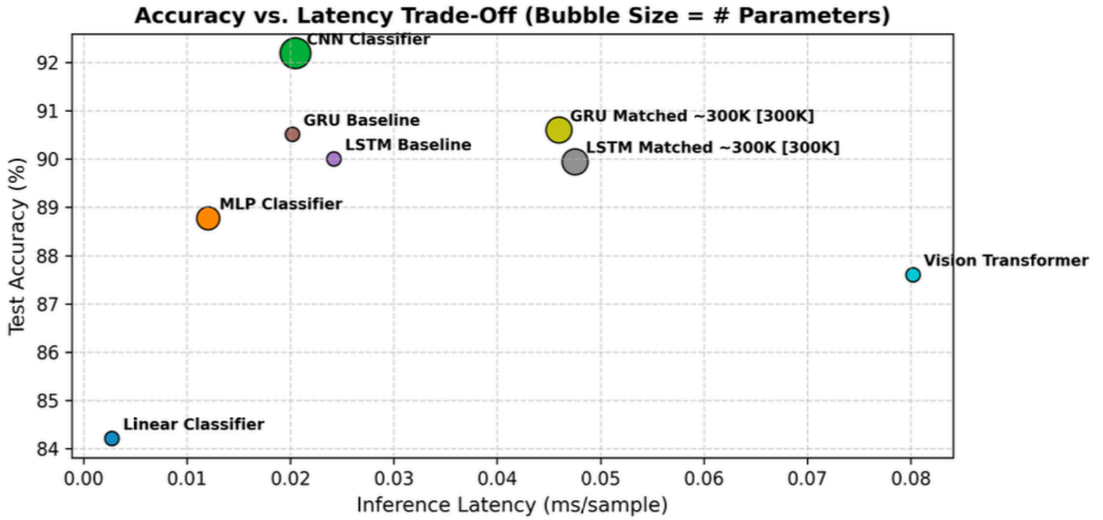
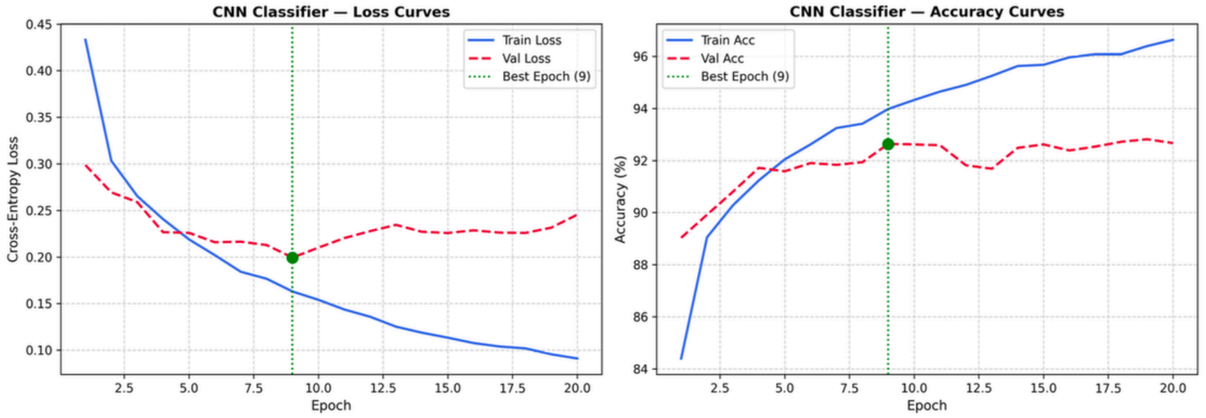


Figure 3.8: Master architectural trade-off analysis: Accuracy vs. Parameter Capacity vs. Latency.

MAIN EXPERIMENTAL RESULTS

(c) CNN Classifier (Best Validation Loss: 0.1994 at Epoch 9)



(d) Vision Transformer (ViT from Scratch, Best Validation Loss: 0.3170 at Epoch 20)

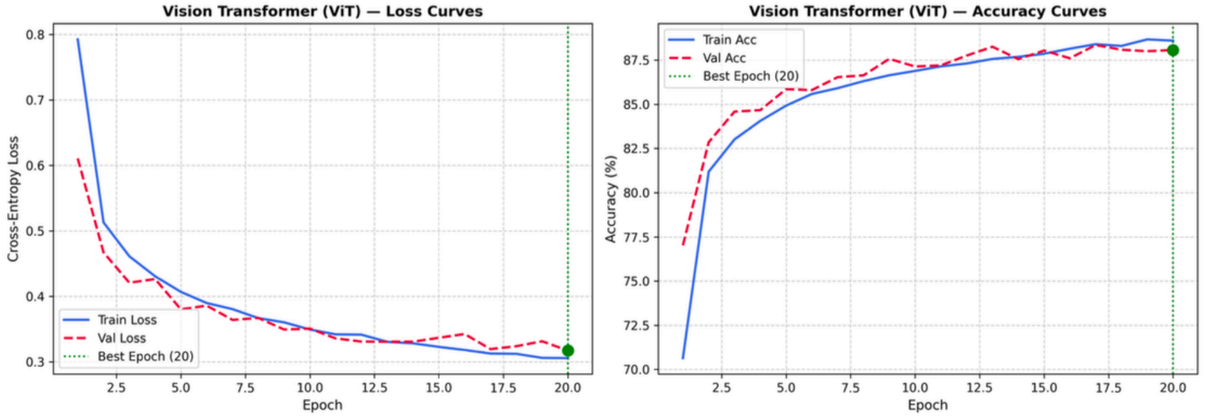
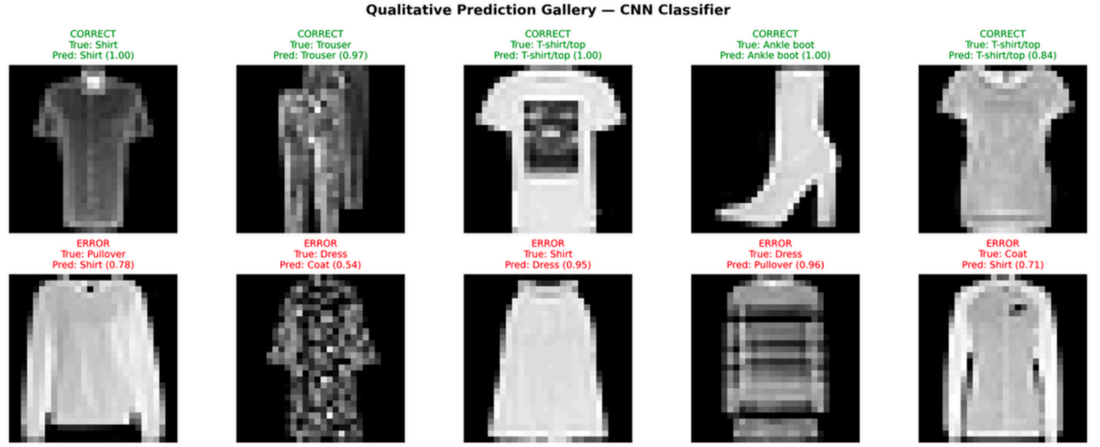


Figure 3.2: Training and validation loss and accuracy curves for CNN and Vision Transformer.

(a) CNN Classifier Predictions (Test Accuracy: 92.19%)



(b) Vision Transformer (ViT from Scratch) Predictions (Test Accuracy: 87.60%)

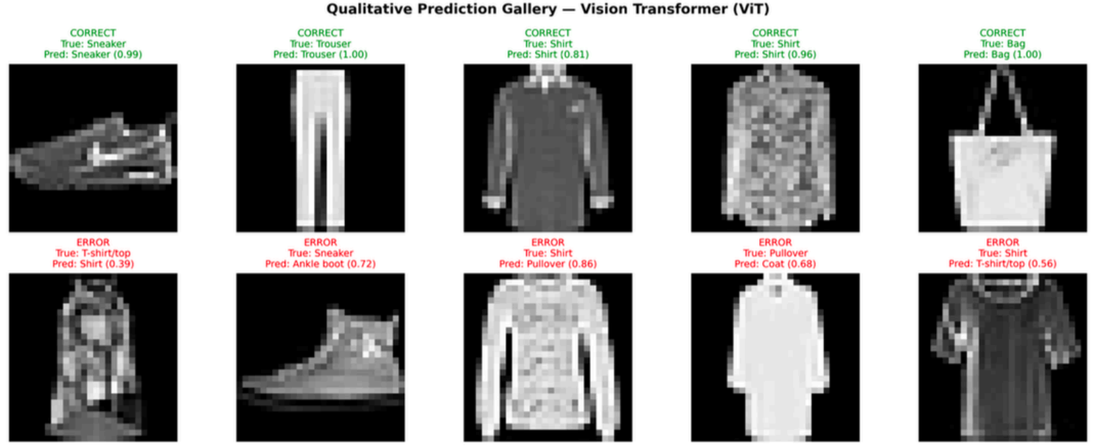
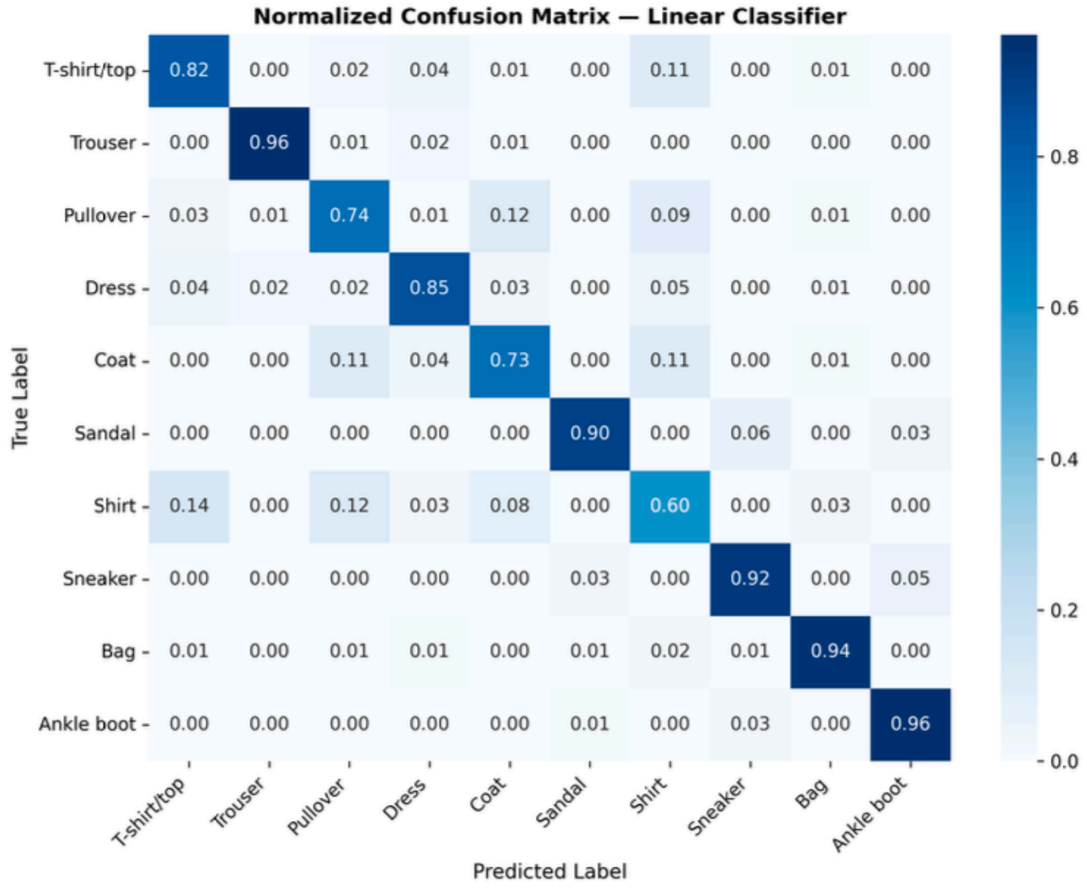


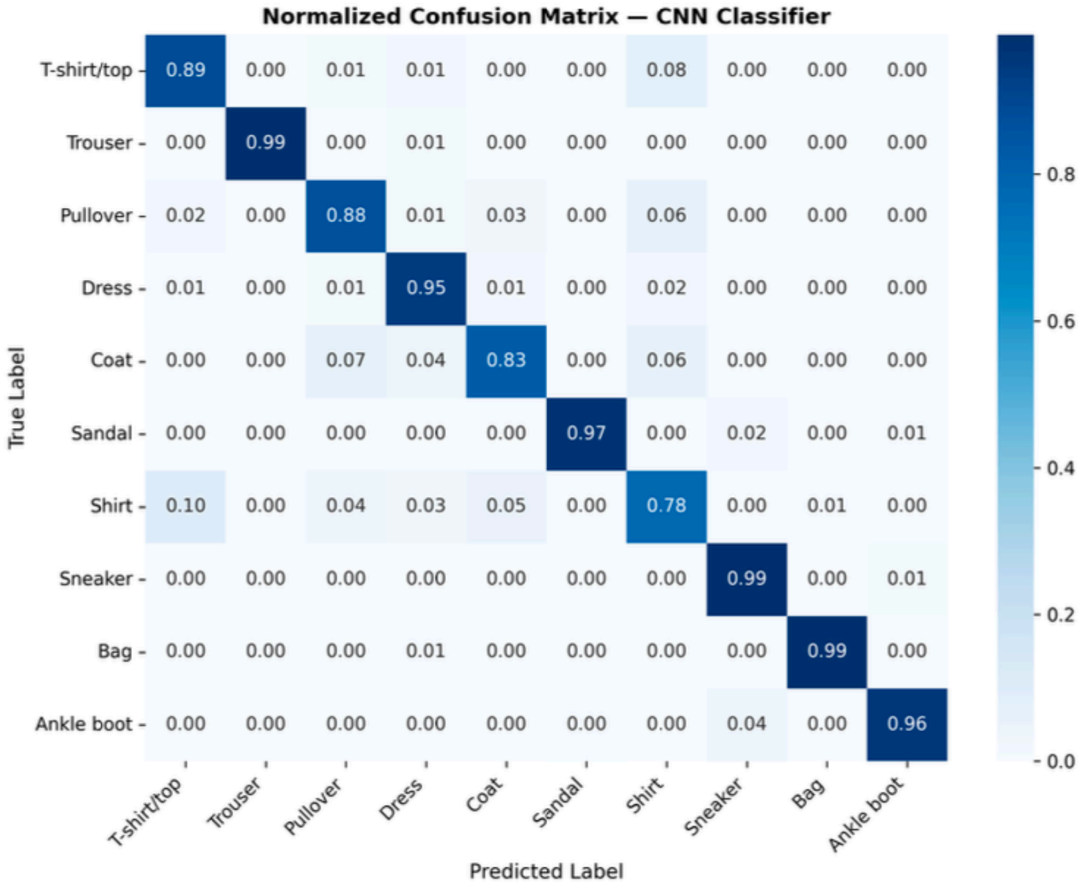
Figure 3.6: Qualitative test sample predictions for CNN and Vision Transformer models.

MAIN EXPERIMENTAL RESULTS

*Normalized **confusion matrices** identifying primary misclassification clusters.*



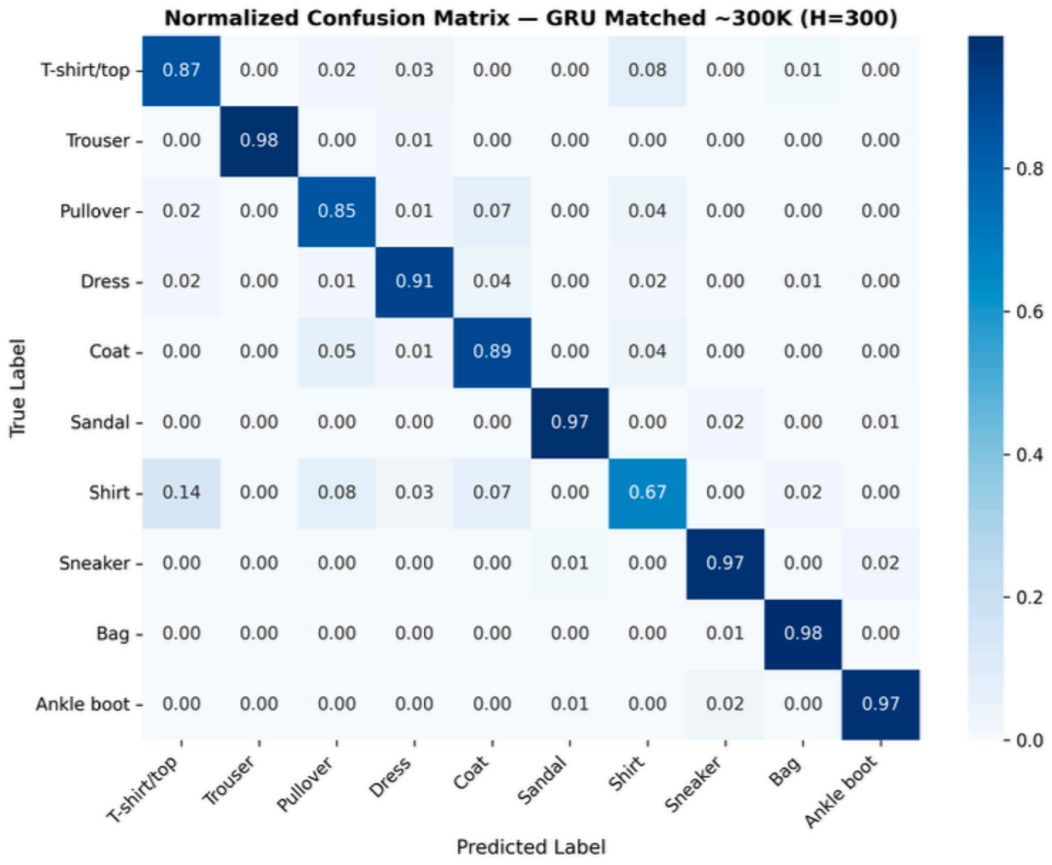
(a) Linear Classifier



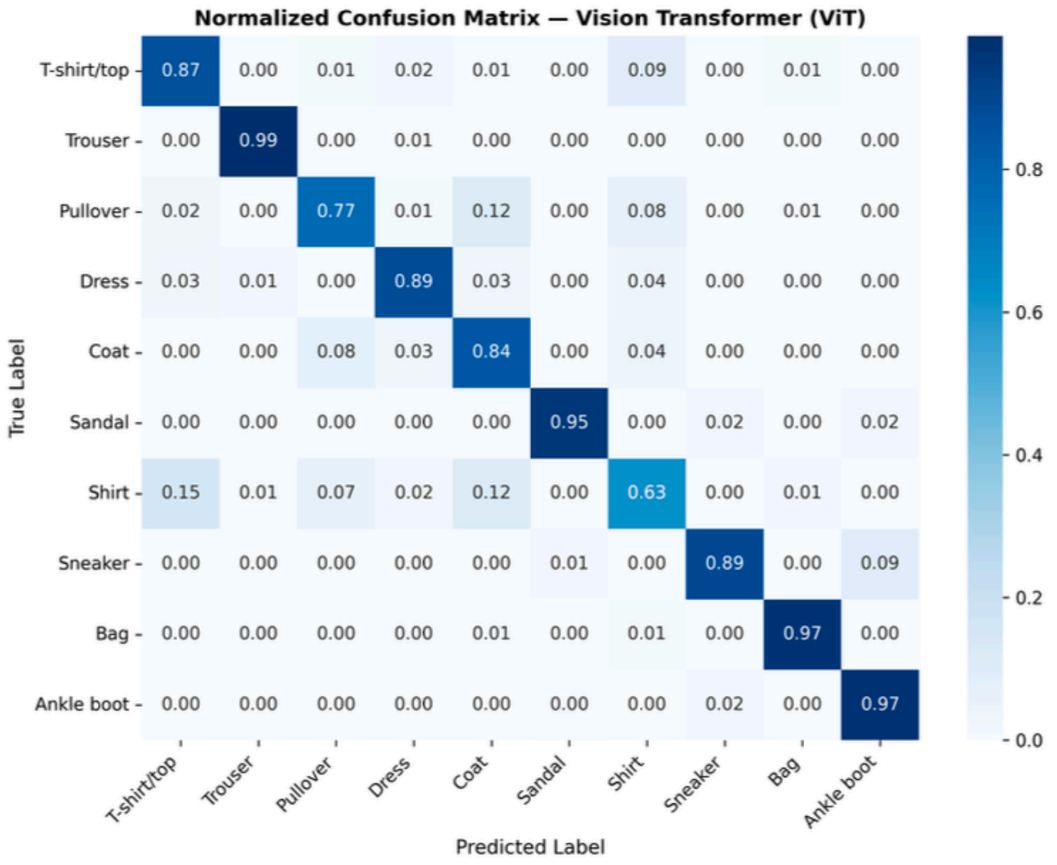
(b) CNN Classifier

MAIN EXPERIMENTAL RESULTS

*Normalized **confusion matrices** identifying primary misclassification clusters.*



(c) GRU Matched ($H = 300$)



(d) Vision Transformer

MAIN EXPERIMENTAL RESULTS

ERROR ANALYSIS

***Key Finding**: Errors consistently cluster around visually similar fashion categories.*

- **Upper Garments** (Shirt ↔ T-shirt/top): Most frequent confusion across all models (97–149 errors), caused by lost fine-grained details.
- **Outerwear** (Pullover ↔ Coat): Similar silhouettes and long sleeves lead to misclassification.
- **Footwear** (Sneaker ↔ Ankle Boot): Overlapping shapes and shaft heights cause confusion (93 errors in ViT).

Conclusion: Low image resolution (28 × 28) limits fine-grained visual discrimination.

CONCLUSION

Limitations

- Dataset: Limited real-world variability (lighting, occlusion, backgrounds).
- RNN: Row-wise processing loses 2D spatial relationships.
- ViT: Training from scratch without pre-training limits performance.

Key Findings

- CNN: Best overall performance - **92.19% accuracy**, benefiting from spatial inductive bias.
- GRU vs. LSTM: GRU achieves **+0.66% accuracy** with matched parameters (~300K) and converges 4 epochs faster.

Thank you for listening